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min $3y_1 + 4y_2 + 5y_3$ $y_1 + y_2 \geq 1$ $y_2 + y_3 \geq 1$ $y_1 + y_3 \geq$

a) min $c^T y$ при $Ay \geq b$, $0 \leq y \leq 1$

$c = (3, 4, 5)^T$

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix} \quad b^T = (1, 1, 1)$$

b) $-Ay \leq -b$ change to the opposite sign

$$A' = \begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & -1 \\ -1 & 0 & -1 \end{pmatrix} \quad b' = \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}$$

c) ~~Plot~~

(y_1, y_2, y_3)	$y_1 + y_2 \geq 1$	$y_2 + y_3 \geq 1$	$y_1 + y_3 \geq 1$	accept	$Z = 3y_1 + 4y_2 + 5y_3$
(0, 0, 0)	X	X	X	no	0
(0, 0, 1)	X	V	V	no	5
(0, 1, 0)	V	V	X	no	4
(0, 1, 1)	V	V	V	yes	9
(1, 0, 0)	V	X	V	no	3
(1, 0, 1)	V	V	V	yes	8
(1, 1, 0)	V	V	V	yes	7
(1, 1, 1)	V	V	V	yes	12

Four acceptable points and min equals 60 7

$z^* = 7 \quad y^* = (1, 1, 0)$